MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER PRACTICAL EXAMINATION

(2024 ADMISSION ONWARDS)

Model Question Paper

MG3DSCECT203 INTEGRATED ELECTRONICS

Duration: 2 hours **Maximum Marks: 35**

Instructions:

- **Demonstrate** the given task
- **Viva session** (Minimum 5 Questions from the practical module)
- **Certified Lab Report**

Evaluation Criteria:

- **Viva:** 7 Marks
- **Lab Report :** 8 Marks
- **Demonstration:** 20 Marks

Tasks:

1. Set up a Zero crossing detector using an Op-amp and plot its output waveform
2. Set up an inverting amplifier for gain 2 using Op-amp with different input voltages and calculate its Practical gain.
3. Set up a non- inverting amplifier for gain 2 using Op-amp with different input voltages and calculate its Practical gain.
4. Construct an Astable multivibrator using a 555 timer and plot the output waveform.
5. Construct a Decade counter using IC and verify the truth table.
6. Construct a Ring counter using IC and verify the truth table.
7. Construct a Johnson counter using IC and verify the truth table.
8. Construct a Ripple counter using IC and verify the truth table.
9. Set up a Parallel in Parallel out shift register using ICs and verify its truth table.
10. Set up a Parallel in Serial out shift register using ICs and verify its truth table.
11. Construct a voltage comparator using Op-amp, with a 10Vpp sinusoidal input and a reference voltage of 1 V and Plot its input-output waveforms.
12. Design an RC coupled amplifier Using BJT and plot its frequency response using a simulation software.
13. Design an RC Phase shift Oscillator Using Op-amp and plot its output waveform using a suitable simulation software.
14. Verify the operation of a Ring and Johnson Counter using simulation tool.
15. Verify the operation of a decade counter Using Simulation tool.